

Gaze-Controlled Pan/Tilt Camera

Steering a physical camera in real time using webcam-based eye tracking — project summary

The Problem

Gaze is one of the fastest, most natural control signals we have, but real eye-tracking hardware costs hundreds to thousands of dollars, putting gaze-based control out of reach for most people, schools, and hobbyists.

The Approach

Built a full pipeline from consumer parts: a webcam + dlib facial landmarks estimate gaze direction on a Raspberry Pi 5, which streams smoothed pan/tilt commands over its own self-hosted Wi-Fi network to a second Pi driving a servo camera mount.

The Result

A working closed loop: look somewhere, and a physical camera turns to follow your gaze in real time — live video and a gaze-debug view streamed back to a browser, all running on ~\$150 of off-the-shelf hardware.

How it works

Base Station — Raspberry Pi 5

Webcam → gaze tracking → servo commands
Hosts Wi-Fi AP for the camera Pi

~\$150

Total hardware cost



Wi-Fi (192.168.4.0/24) — video + control

20 Hz

Servo update rate

Camera Pi

Smooths commands → pan/tilt servos
Streams H.264 video back over Wi-Fi

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Specialized eye-tracking hardware

Why This Matters

Commercial eye trackers cost hundreds to thousands of dollars. This project demonstrates the core idea: detect where someone is looking and move a physical device there, using a webcam and two Raspberry Pis at a fraction of the cost.

Accessibility

People with limited hand or arm mobility — from ALS, spinal cord injuries, or muscular dystrophy — are a primary use case for commercial eye-tracking systems, for controlling a computer, a communication device, or a wheelchair-mounted camera. A low-cost, open version of this pipeline is a step toward making that kind of control cheaper and more accessible.

Robotics & HCI

Gaze is a natural fit for hands-free teleoperation: aiming a camera, a telepresence robot, or a drone gimbal just by looking, freeing the hands for other controls entirely. This project is a working demonstration of that control loop end to end.

Education & Research

Built entirely from open-source tools (dlib, OpenCV) and commodity hardware, the whole pipeline is something a student or hobbyist can inspect, modify, and build on — instead of a closed commercial black box.

The goal isn't this specific rig — it's proving that gaze-based control doesn't have to be expensive or exotic to be functional.